Statistical Mechanics of Reinforcement Learning

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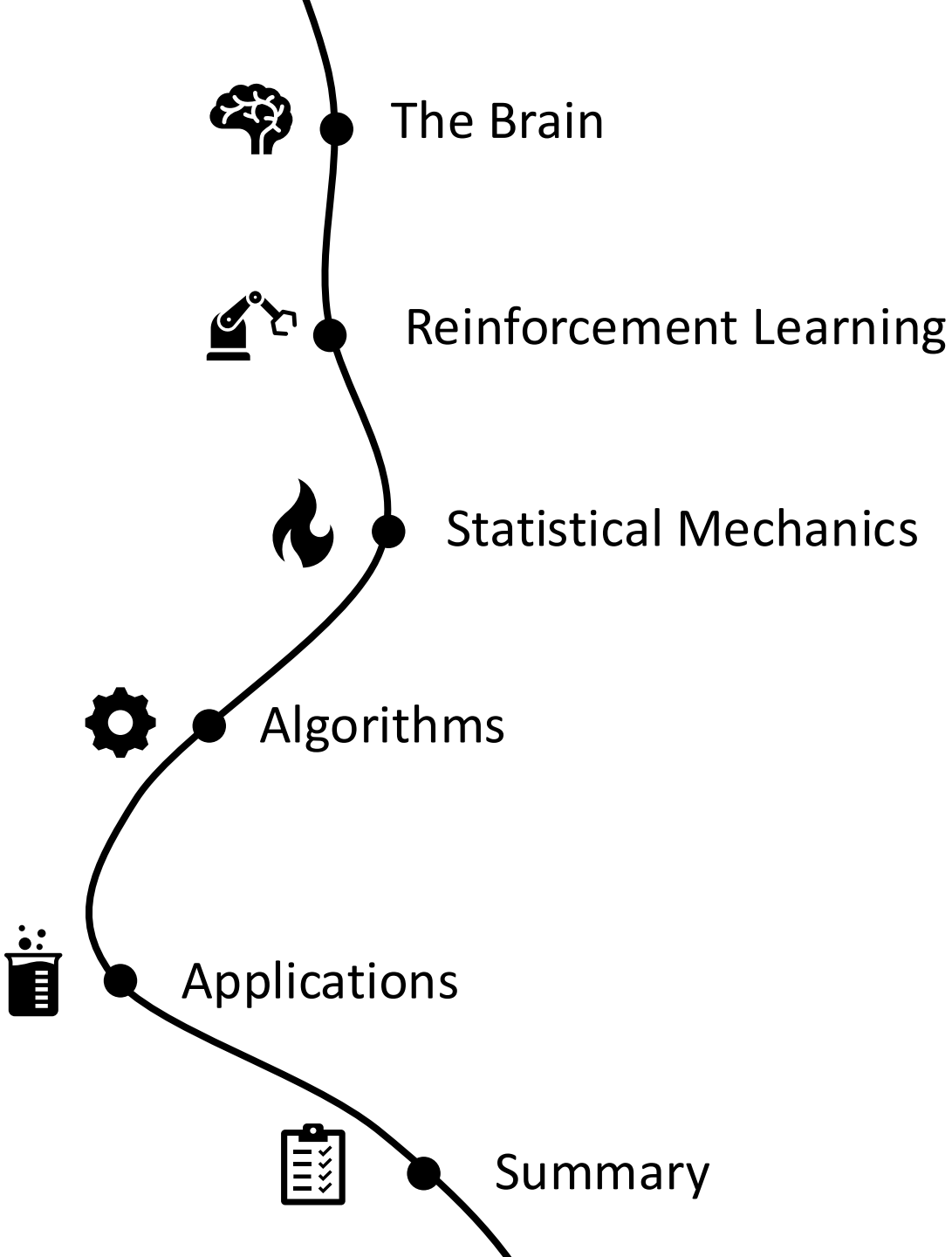
18 March 2026



Follow along with more details and references

The Brain

Reinforcement Learning

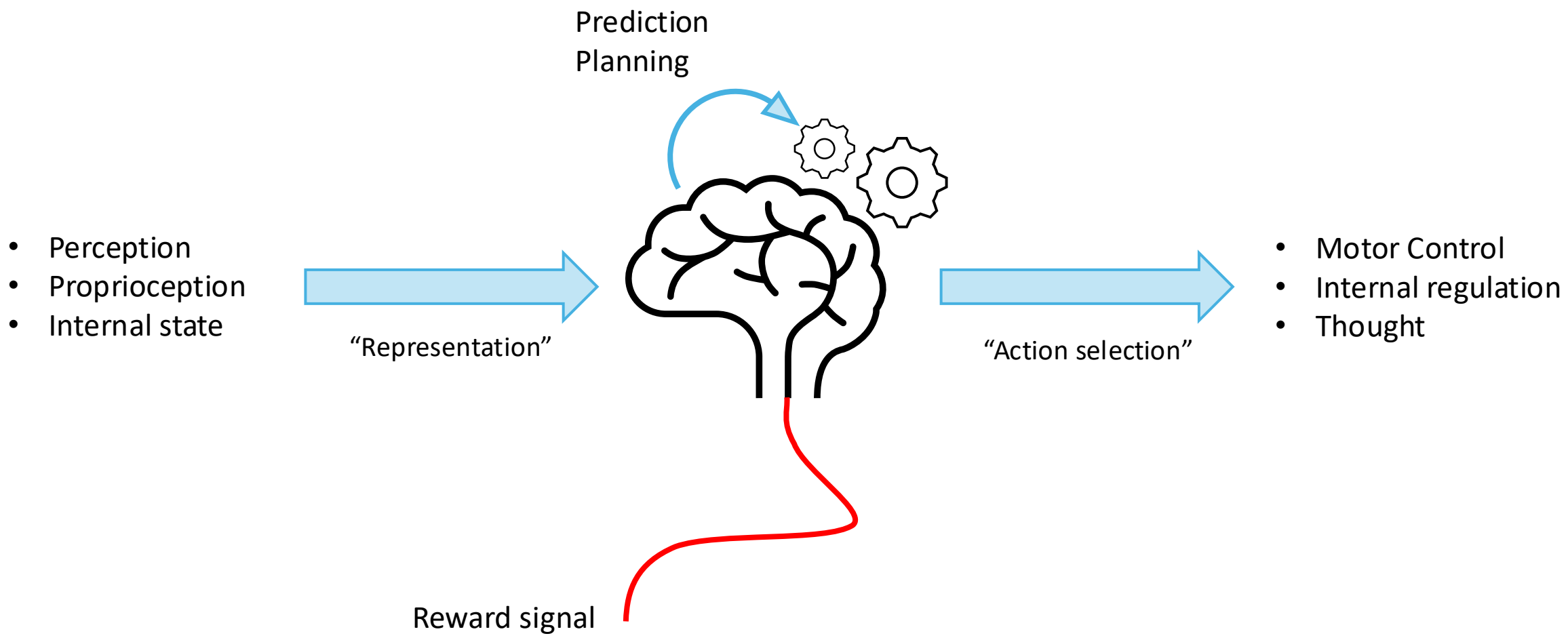
Statistical Mechanics

Algorithms

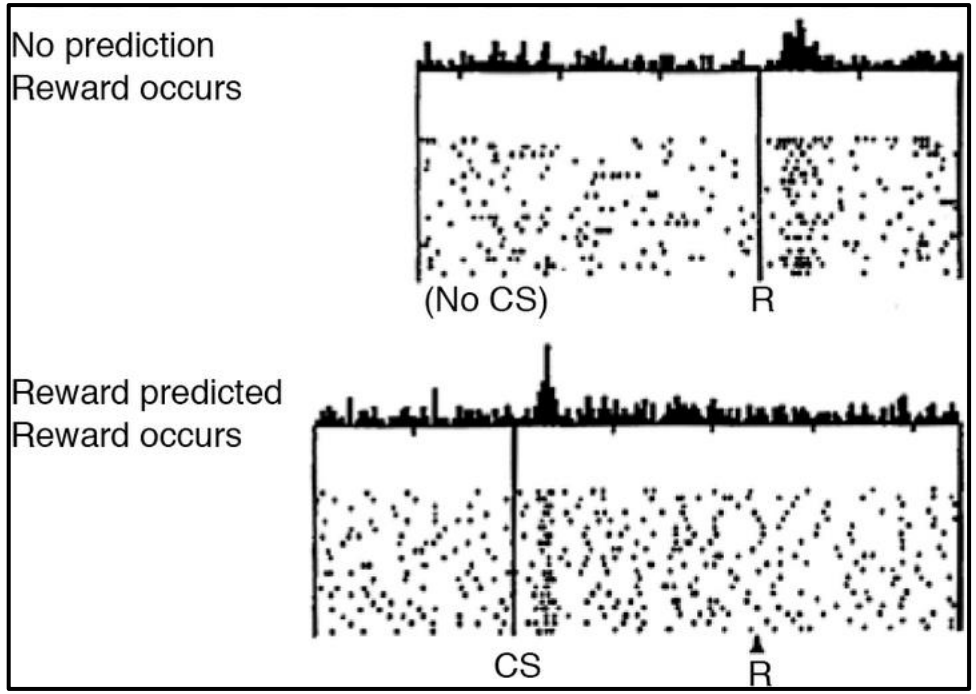
Applications

Summary

The brain as a prediction engine



Perceiving and Acting



Reinforcement learning in the brain

Yael Niv
Psychology Department & Princeton Neuroscience Institute, Princeton University

Abstract: A wealth of research focuses on the decision-making processes that animals and humans employ when selecting actions in the face of reward and punishment. Initially such work stemmed from psychological investigations of conditioned behavior, and explanations of these in terms of computational models. Increasingly,

A Neural Substrate of Prediction and Reward

Wolfram Schultz, Peter Dayan, P. Read Montague*

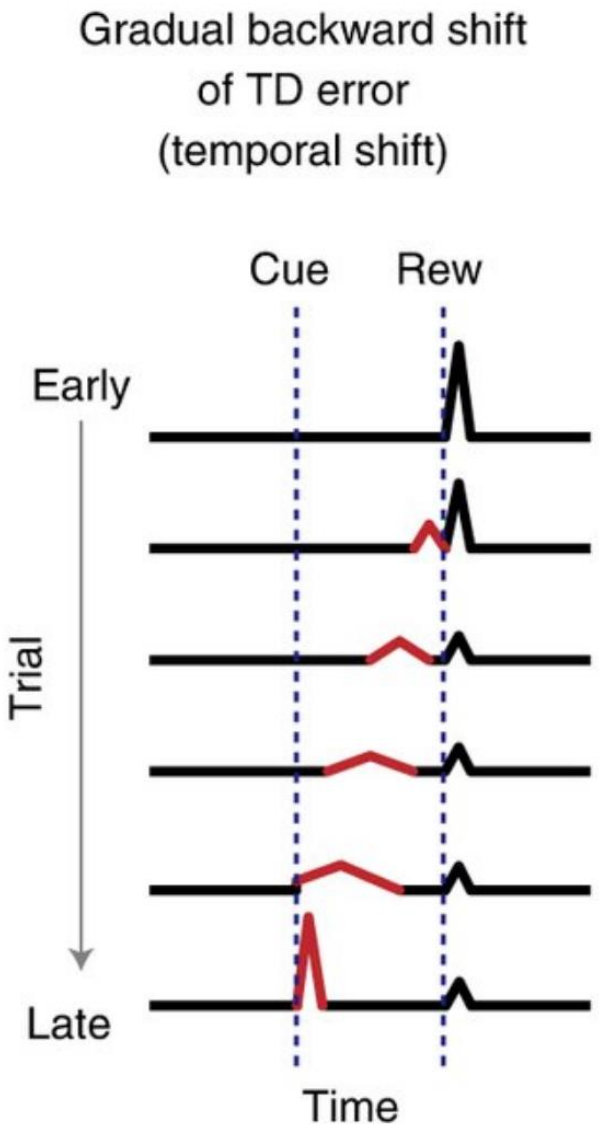
The capacity to predict future events permits a creature to detect, model, and manipulate the causal structure of its interactions with its environment. Behavioral experiments suggest that learning is driven by changes in the expectations about future salient events such as rewards and punishments. Physiological work has recently complemented these studies by identifying dopaminergic neurons in the primate whose fluctuating output apparently signals changes or errors in the predictions of future salient and rewarding events. Taken together, these findings can be understood through quantitative theories of adaptive optimizing control.

This dopaminergic neuron firing (and its adaptation) is explained by the temporal-difference error

Temporal differences

- Eligibility traces
- Prediction error
- Credit assignment
- Optimal control

One signal drives learning



Nature Neuroscience 2022, Amo et al.

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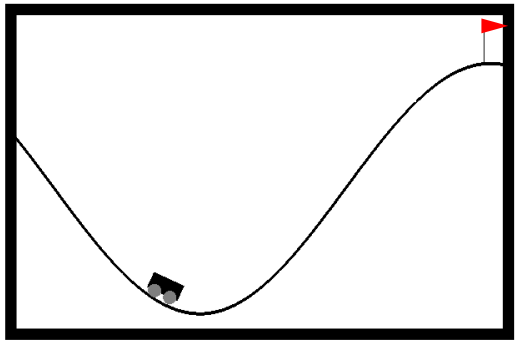
Reinforcement Learning



Reinforcement Learning (RL) is a paradigm created to solve sequential decision-making problems

Basic Ideas

- An agent interacts with the environment (collects its own data)
- Positive behaviors are reinforced relative to undesirable behaviors
  - Reinforcement is implemented via a reward function
- After many interaction-reinforcement cycles, the agent should learn to "successfully" interact with the environment

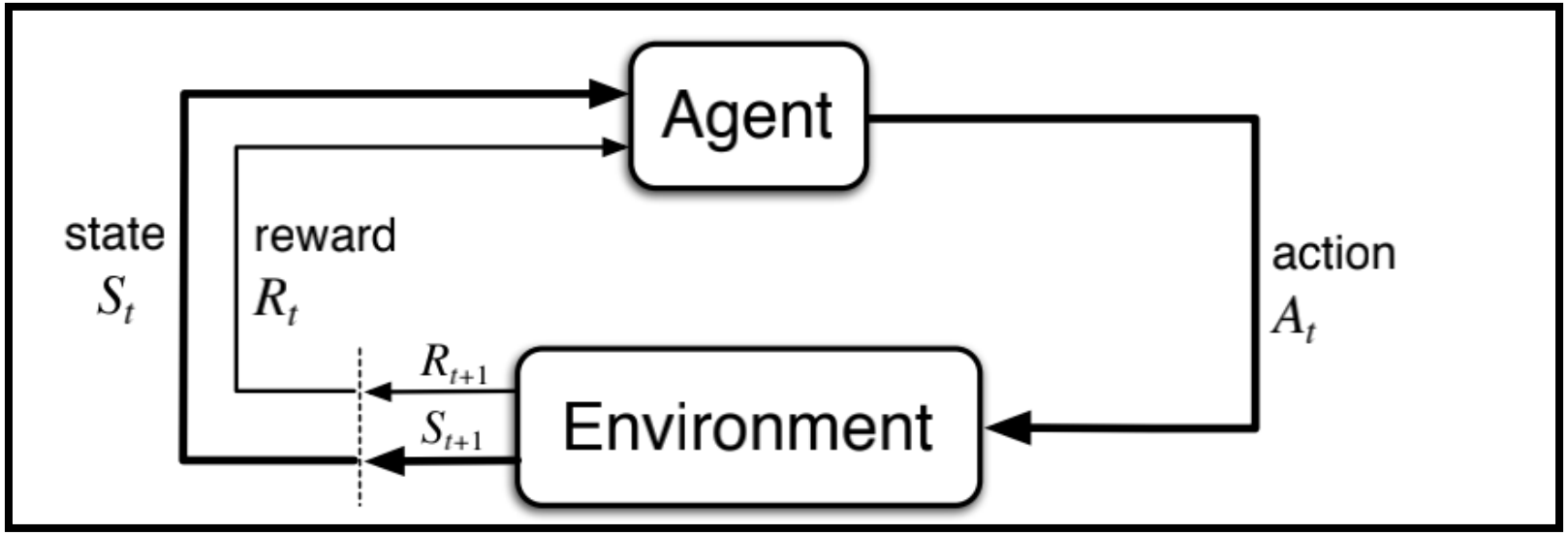
Success is determined by accumulation of reward

Markov Decision Process

- The <u>state</u> (s) and <u>action</u> (a) label the agent's steps in a <u>trajectory</u>:

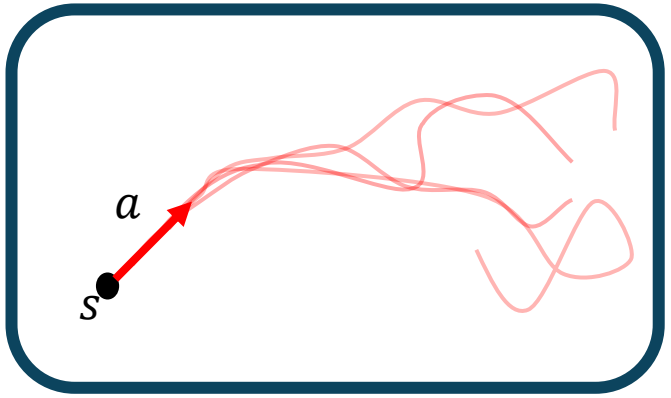
$$\tau = (s_1, a_1, s_2, a_2, \dots)$$

- Transition dynamics are Markovian
  - Dynamics (p) can be either stochastic or deterministic
- Want a <u>policy</u> $\pi(a|s)$ which chooses actions that maximize total reward



Sutton, R. S.; and Barto, A. G. 2018. Reinforcement learning: An introduction. MIT press

Solving the RL Problem



Trajectories induced by π, p; given an initial state and action

<u>What does the solution look like?</u>

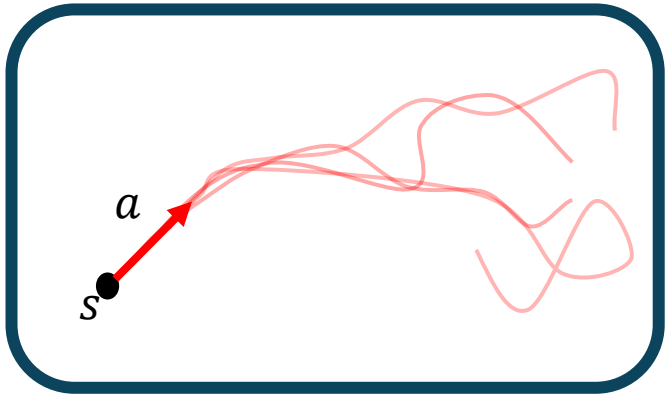
We need to estimate the value of a given decision-making strategy (policy)

$$\rho^{\pi} = \lim_{T \to \infty} \frac{1}{T} \mathbb{E}_{\tau \sim p, \pi} \sum_{t=1}^{\infty} r(s_t, a_t)$$

Average reward rate
(energy density, dopamine baseline)

Number of timsteps in trajectory

Solving the RL Problem



Trajectories induced by π, p; given an initial state and action

<u>What does the solution look like?</u>

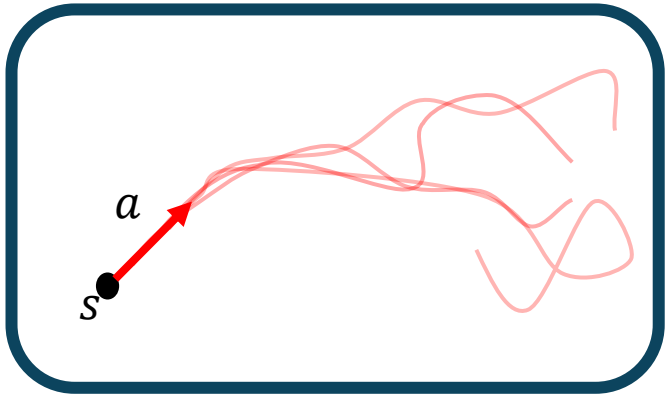
We need to know the decision-making strategy (policy) which attains the highest expected value.

Attribute a <u>value</u> for every (initial) state and action

$$q^{\pi}(s,a) = \lim_{T\to\infty} \mathbb{E}_{\tau\sim p,\pi} \sum_{t=1}^{\infty} \left(r(s_t,a_t) - \rho^{\pi}\right)$$

Differential value function
(relative advantage)

Solving the RL Problem



Trajectories induced by π, p; given an initial state and action

<u>What does the solution look like?</u>

We need to know the decision-making strategy (policy) which attains the highest expected value.

Formulate the following objective function:

$$\rho^* = \max_{\pi} \lim_{T \to \infty} \frac{1}{T} \mathbb{E}_{\tau \sim p, \pi} \sum_{t=1}^{\infty} r(s_t, a_t)$$

Average reward rate
(energy density, dopamine baseline)

049 2 1

Why RL?

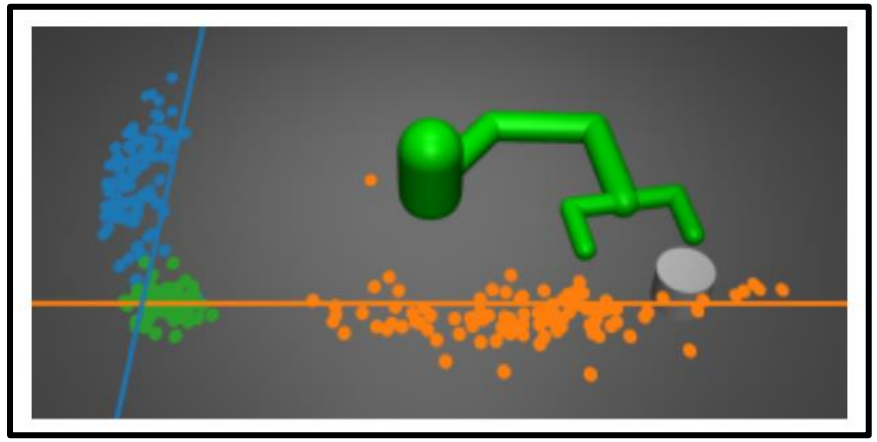
Better describes experimental data!

New formalisms, concepts, etc. can drive new hypotheses for how the brain works!

- Compositionality
- Distributional RL
- Reward Shaping
- Average-Reward

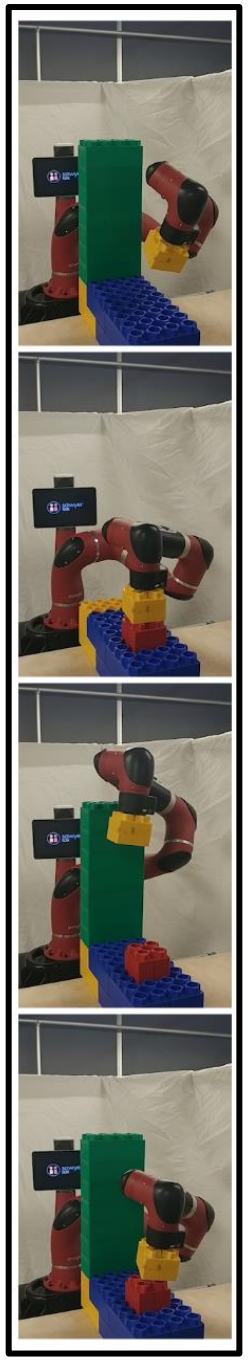
Why RL? (Compositionality)

$$Q^*_{\mathcal{C}}(\mathbf{s},\mathbf{a}) \approx Q_{\Sigma}(\mathbf{s},\mathbf{a}) = \frac{1}{|\mathcal{C}|}\sum_{i\in\mathcal{C}} Q^*_i(\mathbf{s},\mathbf{a})$$



Composable Deep Reinforcement Learning for Robotic Manipulation

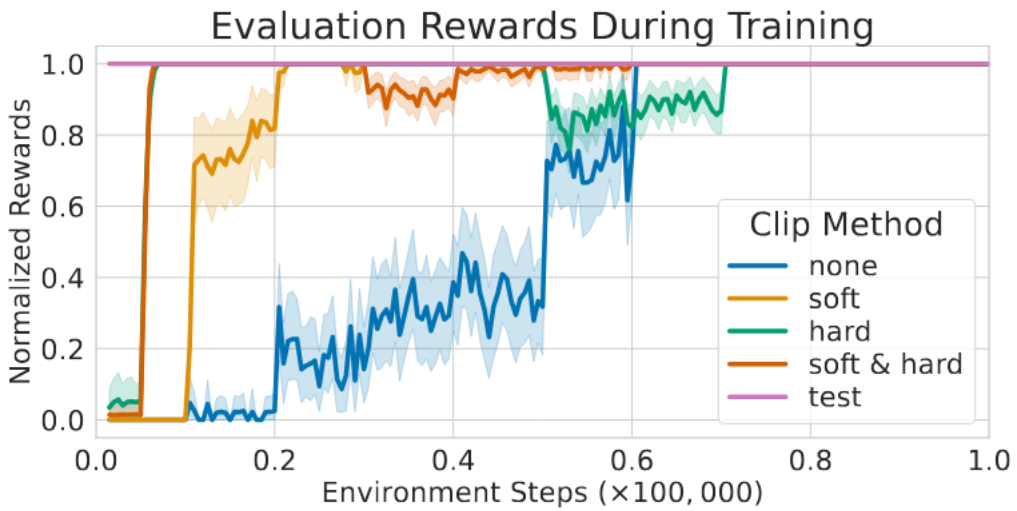
Tuomas Haarnoja[1], Vitchyr Pong[1], Aurick Zhou[1], Murtaza Dalal[1], Pieter Abbeel[1,2], Sergey Levine[1]

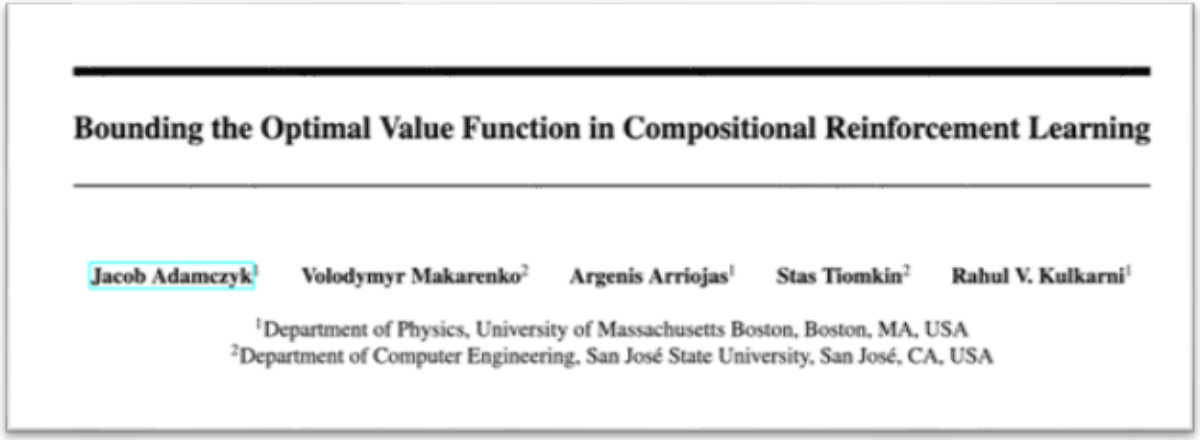


Why RL? (Compositionality)

$$Q^*_{\mathcal{C}}(\mathbf{s},\mathbf{a}) \approx Q_{\Sigma}(\mathbf{s},\mathbf{a}) = \frac{1}{|\mathcal{C}|}\sum_{i\in\mathcal{C}} Q^*_i(\mathbf{s},\mathbf{a})$$

Standard RL Results	
Transformation	Result
Linear Map:	$\widetilde{Q}(s,a) = kQ(s,a)$
Convex conditions:	$\widetilde{Q}(s,a) \geq f(Q(s,a))$
Concave conditions:	$\widetilde{Q}(s,a) \leq f(Q(s,a))$
OR Composition:	$\widetilde{Q}(s,a) \geq \max_k\{Q^{(k)}(s,a)\}$
AND Composition:	$\widetilde{Q}(s,a) \leq \min_k\{Q^{(k)}(s,a)\}$
NOT Gate:	$\widetilde{Q}(s,a) \geq -Q(s,a)$
Conical combination:	$\widetilde{Q}(s,a) \leq \sum_k \alpha_k Q^{(k)}(s,a)$

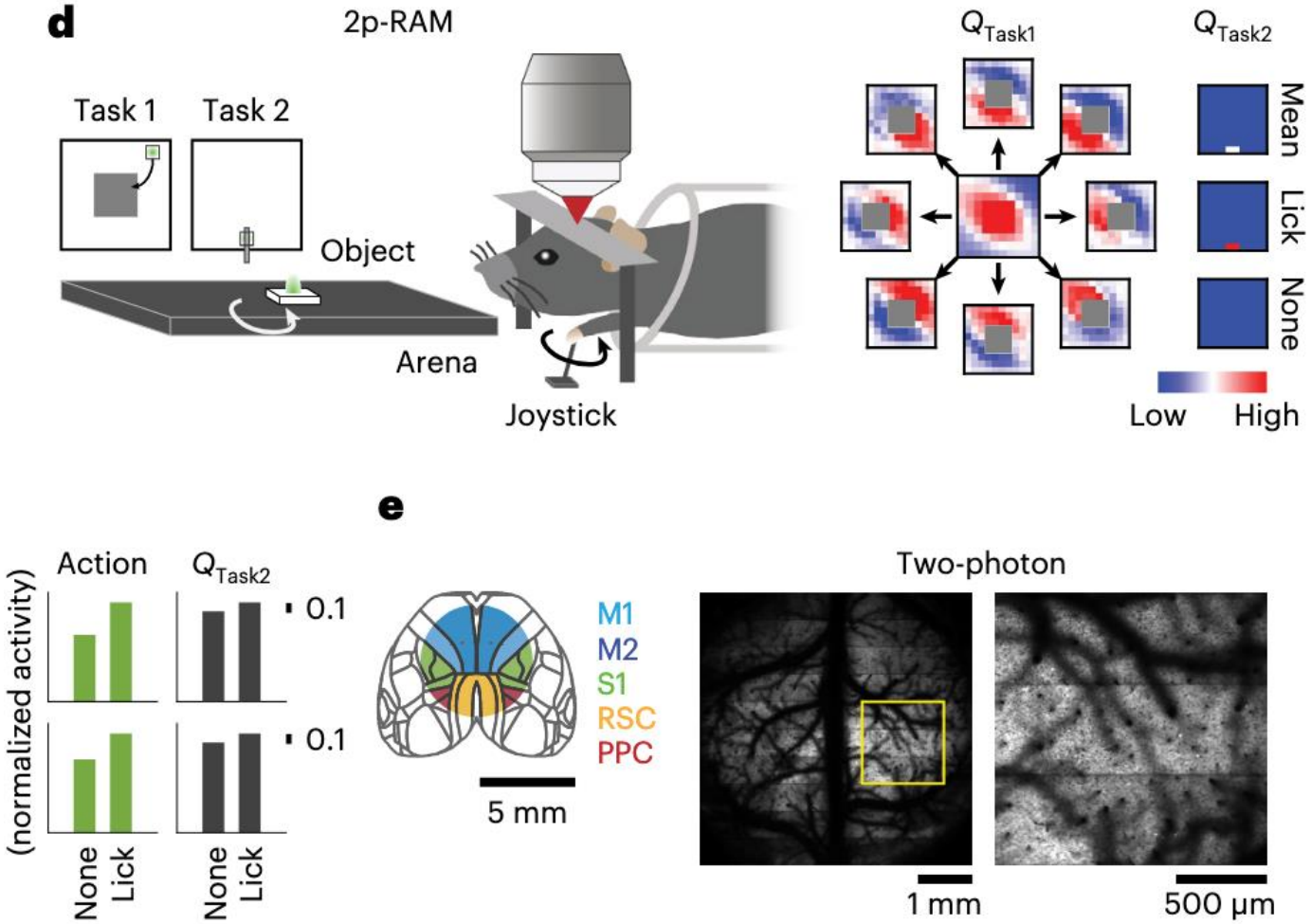


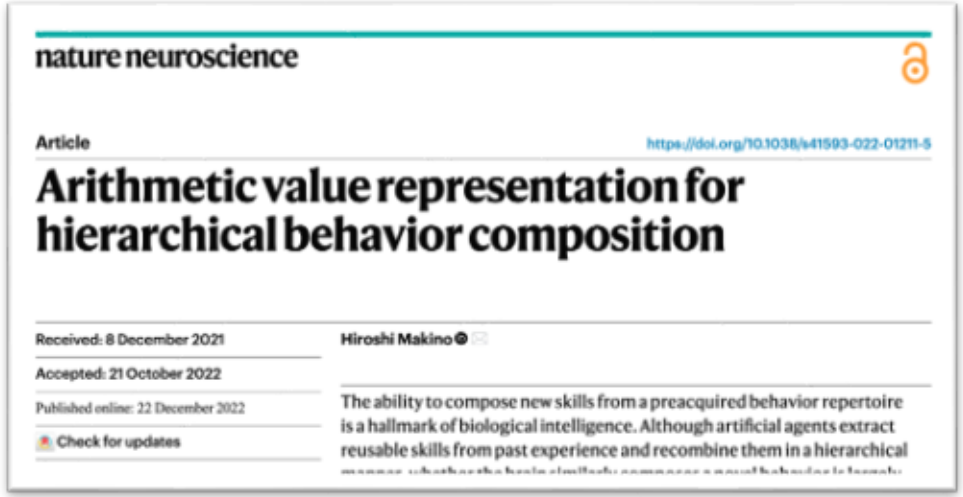

Bounding the Optimal Value Function in Compositional Reinforcement Learning

Jacob Adamczyk[1] Volodymyr Makarenko[2] Argenis Arriojas[1] Stas Tiomkin[2] Rahul V. Kulkarni[1]

[1]Department of Physics, University of Massachusetts Boston, Boston, MA, USA
[2]Department of Computer Engineering, San José State University, San José, CA, USA

Why RL? (Compositionality)




nature neuroscience

Article

https://doi.org/10.1038/s41593-022-01211-5

Arithmetic value representation for hierarchical behavior composition

Received: 8 December 2021

Accepted: 21 October 2022

Published online: 22 December 2022

Check for updates

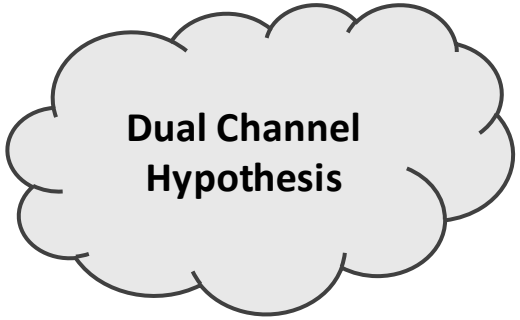
Hiroshi Makino

The ability to compose new skills from a preacquired behavior repertoire is a hallmark of biological intelligence. Although artificial agents extract reusable skills from past experience and recombine them in a hierarchical

Why RL? (Shaping)

- “How” Rewards vs. “What” Rewards
- A model of bias (initialization)
- Preferences and hints that do not affect optimality

Why RL? (Average-Reward)



Phasic and tonic dopamine

Reward prediction error

Reward-rate

The same RPE updates both channels!

"…as the reward rate increases, the amplitude of phasic responses to unexpected rewards should become progressively smaller, as observed experimentally."

"The average reward model formalizes the longstanding idea that tonic dopamine activity can exert an antagonistic effect on phasic activity through the action of autoreceptors."

Why RL? (Average-Reward)

Dual Channel Hypothesis

Phasic and tonic dopamine

Reward prediction error

Average-reward rate

nature neuroscience

Perspective

https://doi.org/10.1038/s41593-024-01705-4

Explaining dopamine through prediction errors and beyond

Received: 24 April 2023

Accepted: 13 June 2024

Samuel J. Gershman[1,2], John A. Assad[3], Sandeep Robert Datta[3], Scott W. Linderman[4], Bernardo L. Sabatini[2,3,5], Naoshige Uchida[6] & Linda Wilbrecht[7]

The Brain

Reinforcement Learning

Statistical Mechanics

Algorithms

Applications

Summary

Statistical Mechanics Framework

- Can we model this learning system with stat mech?
- What is the large parameter?
- What is the microstate?

Why? We get to leverage fundamental & active research:

- Ensembles
- Policies, generators
- Large deviations
- Non-equilibrium protocols

Statistical Mechanics Framework

"Trajectory Thermodynamics" (not configurations)

$$\tau = (s_1, a_1, s_2, a_2, \dots)$$

- We can use the entire machinery of stat mech + thermo
- Dynamics (no detailed balance, reversibility)
- Reward (non-differentiable, sparse)
- Everything falls out of this identification:

Trajectories as microstates

Themes

1. We recover known results

2. We give new meaning to hyperparameters
   a) “temperature”
   b) “reset penalty”

3. We derive new algorithms and analytical results
   a) State-of-the-art in some domains
   b) New exploration algorithm
   c) New bounds and equations

Microcanonical Ensemble

$$\Omega(R, k|s) = \int_{\tau_k} \delta\left(\sum_{i=0}^{k-1} r(s_i, a_i) - R\right) d\tau_k$$

One partition function holds many policies!

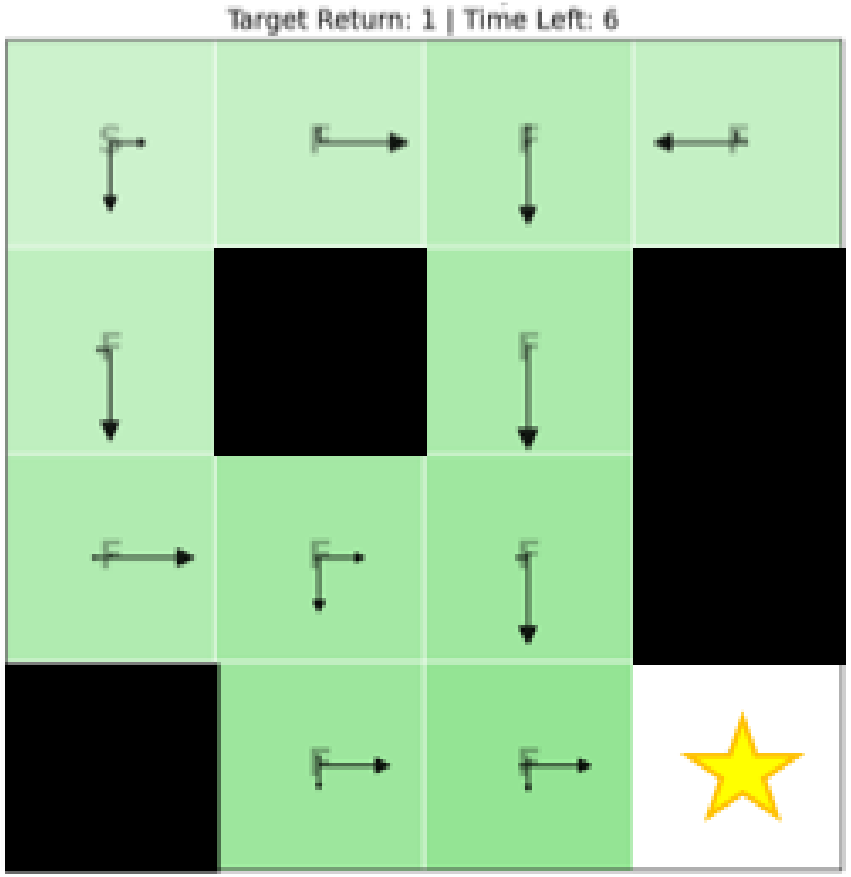
- Decision transformer (DT)
- Upside down RL (UDRL)
- Goal conditioned RL (GCRL)
- Hindsight Experience Replay (HER)

Microcanonical Example

- Reward +1 in bottom right
  - Zero elsewhere
- Start in top left
- Actions: up, down, left right

<u>Request:</u>

- Get a total reward of +1 in 6 steps
- Seeks goal
- MaxEnt with these constraints!

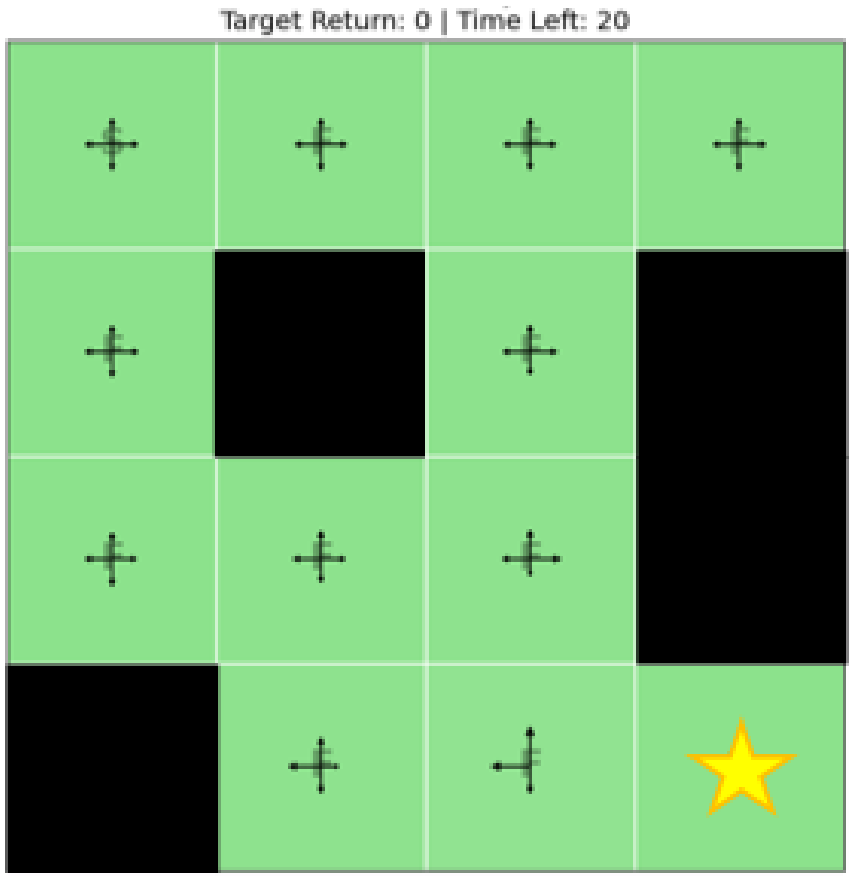


Microcanonical Example

- Reward +1 in bottom right
  - Zero elsewhere
- Start in top left
- Actions: up, down, left right

<u>Request:</u>

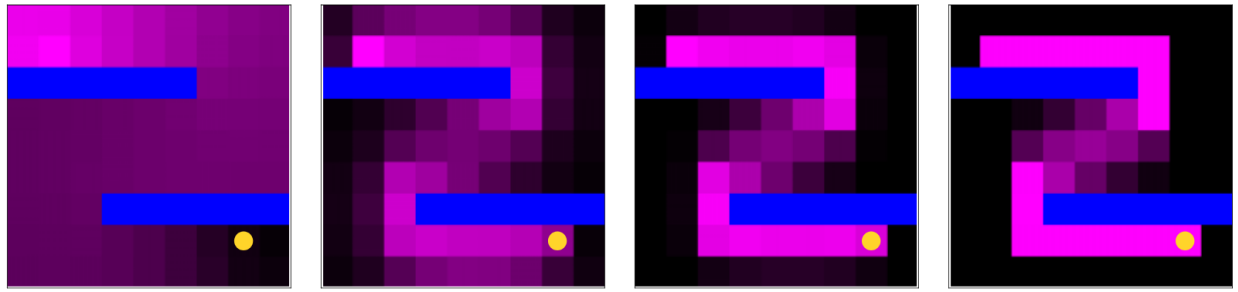
- Get a total reward of 0 in 20 steps
- Avoids goal
- MaxEnt with these constraints!



Canonical Ensemble

$$Z_p(s, a) = \sum_\tau p(\tau|s, a)e^{-\beta E_\tau}$$

- Release the return constraint
- Value function = Free energy
- Recovers MaxEnt RL, soft Q learning (eigenvector equation)
- Physical meaning of beta

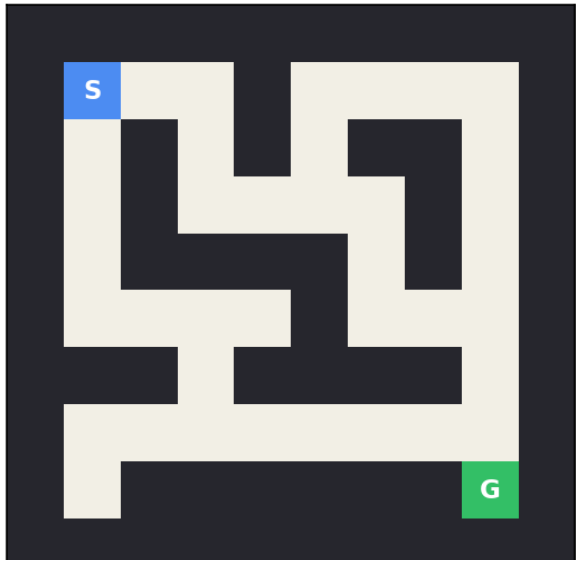


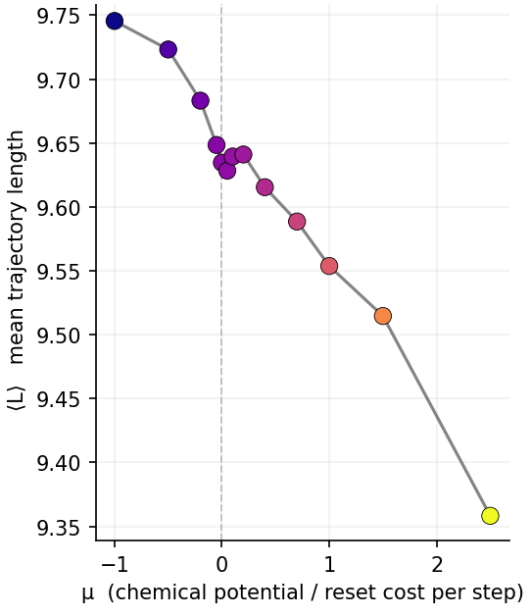
(a) $\beta = 2$ (b) $\beta = 20$ (c) $\beta = 40$ (d) $\beta = 200$

Grand canonical ensemble

$$\Xi(\beta, \mu) = \sum_{\tau} p_0(\tau) e^{\beta(R(\tau)+\mu T)}$$

- Reset Penalty = Chemical potential
- We can control average trajectory length





Give physical meaning to hyperparameters

Thermodynamics

- Equivalence of ensembles ✅
- Maxwell relations ✅
- Minimum work protocols ✅
- Multi-agent systems and mean-field theory 🧐
- Renormalization group flow (Hierarchical RL) 🧐

Hypotheses for the Brain

RL has consistently sprung surprises, so novel ideas are at least worth considering in neuroscience…

Let’s highlight a few!

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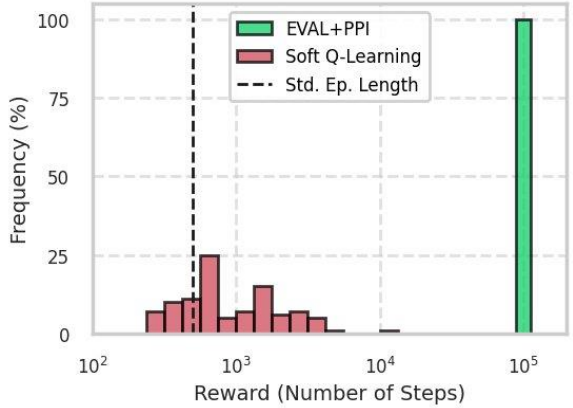
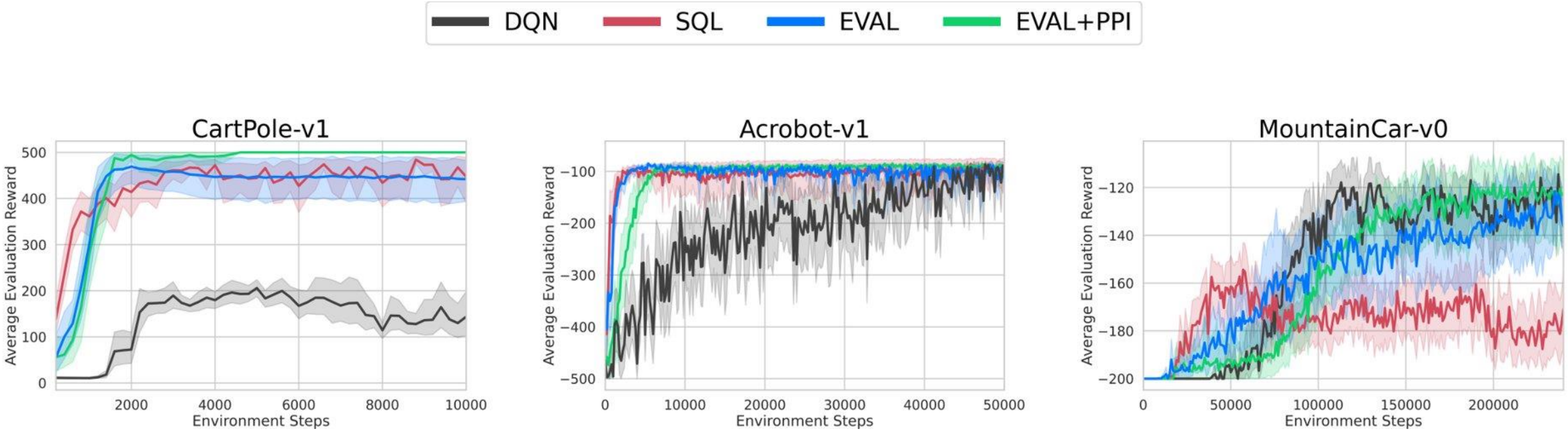
EVAL + PPI



- We can <u>learn</u> the left eigenvector from data
  - Provides optimal policy for entropy-regularized problem
- Posterior Policy Iteration (PPI)



EVAL + PPI

Average-Reward Reinforcement Learning with Entropy Regularization

Jacob Adamczyk[1,2], Volodymyr Makarenko[3], Stas Tiomkin[4], Rahul V. Kulkarni[1,2]
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[2]The NSF Institute for Artificial Intelligence and Fundamental Interactions
[3]Department of Computer Engineering, San José State University
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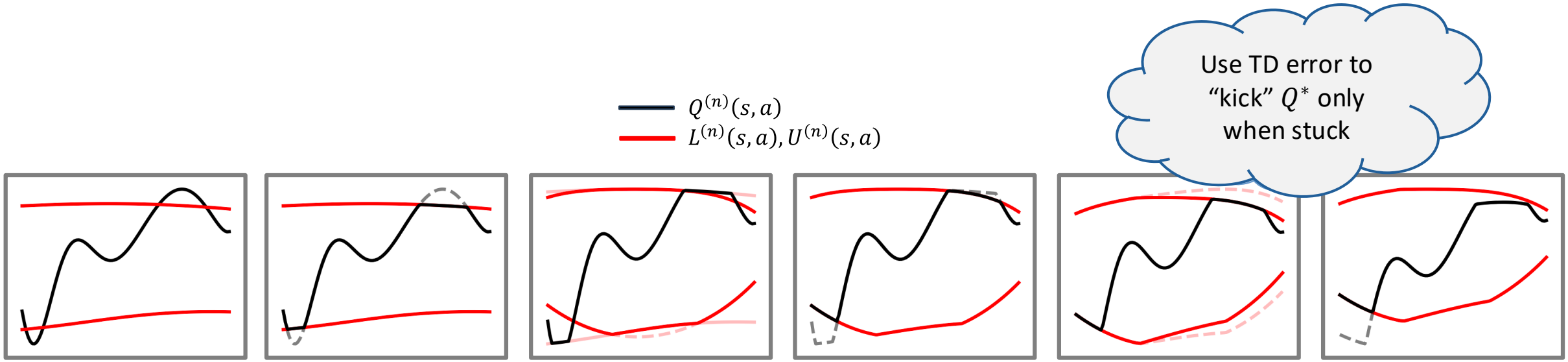
- We can <u>learn</u> the left eigenvector from data
  - Provides optimal policy for entropy-regularized problem
- Posterior Policy Iteration (PPI)

New way to obtain ground state

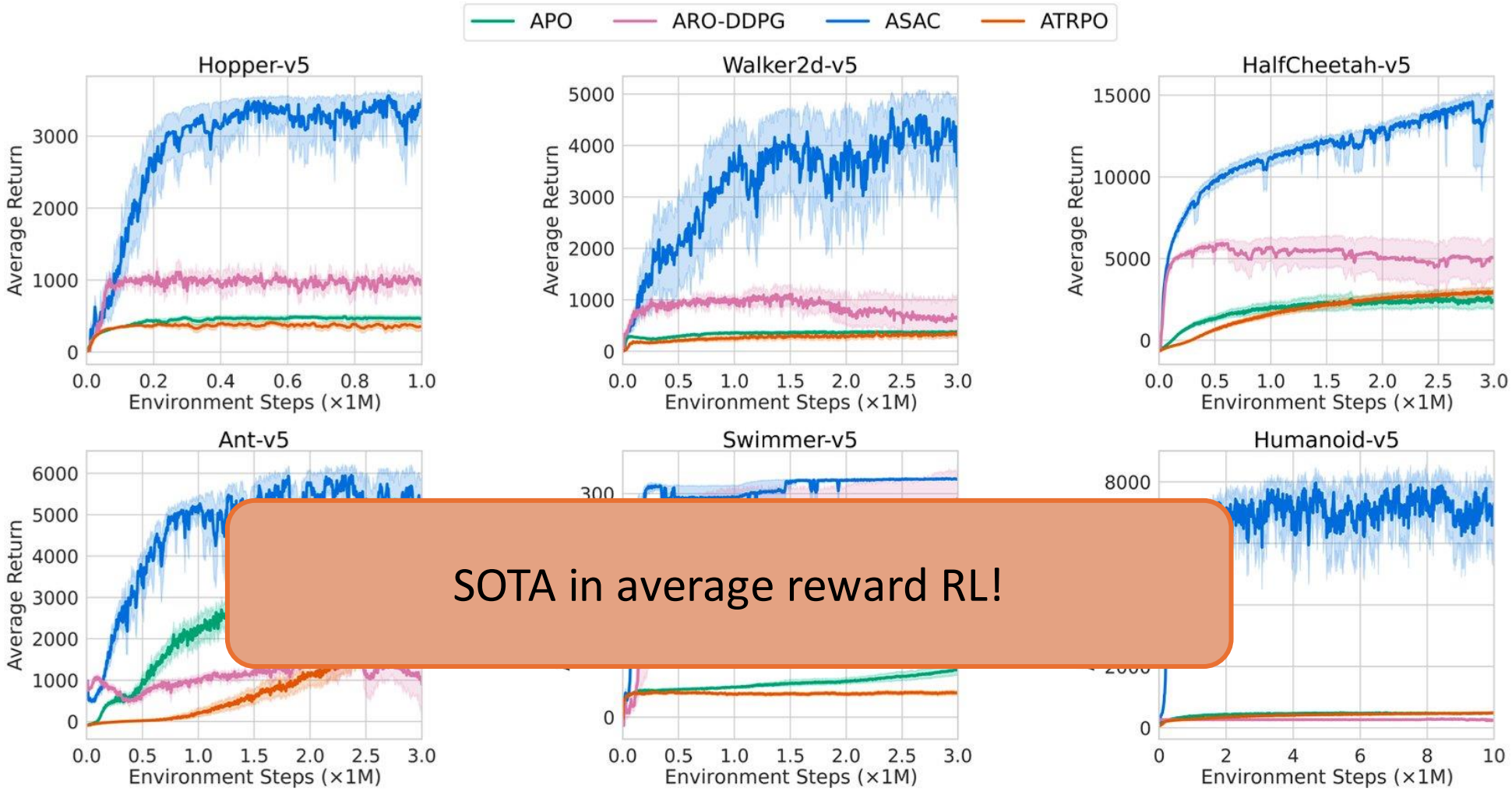
Learning via clipping (Bogoliubov)



Clipping excludes invalid Q^*,
whereas Bellman pulls you toward Q^*

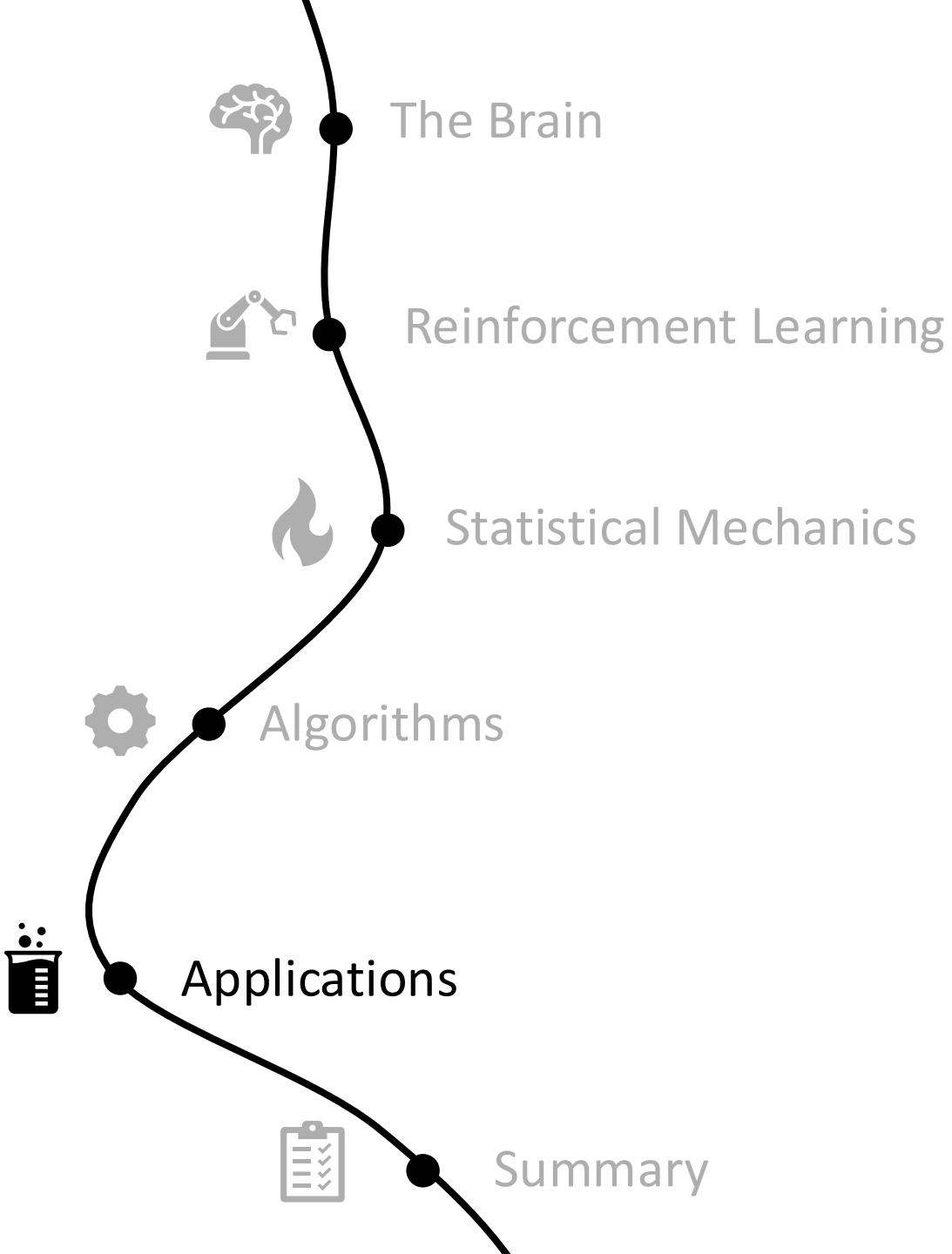
JA, Volodymyr Makarenko, Stas Tiomkin, and Rahul V Kulkarni. "Boosting Soft Q-Learning by Bounding." RLC 2024

ASAC (actor and critic)



SOTA in average reward RL!

JA, Volodymyr Makarenko, Stas Tiomkin, and Rahul V Kulkarni. ”Average-Reward Soft Actor Critic.” RLC 2025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Reward Shaping

What changes to the reward leave the optimal behavior unchanged?

Invariance under gauge transformation

Discrete time gradient!

$$\bar{r}(s,a,s') = \widetilde{r}(s,a,s') + \overbrace{\gamma\Phi(s') - \Phi(s)}$$

$$\bar{Q}^*(s,a) = \widetilde{Q}^*(s,a) - \Phi(s)$$

Choice of gauge (potential)

Reward Shaping

A new dynamical “self-consistent” gauge:

$$\Phi^{(n-1)}(s) = \eta \max_a Q^{(n-1)}(s,a)$$

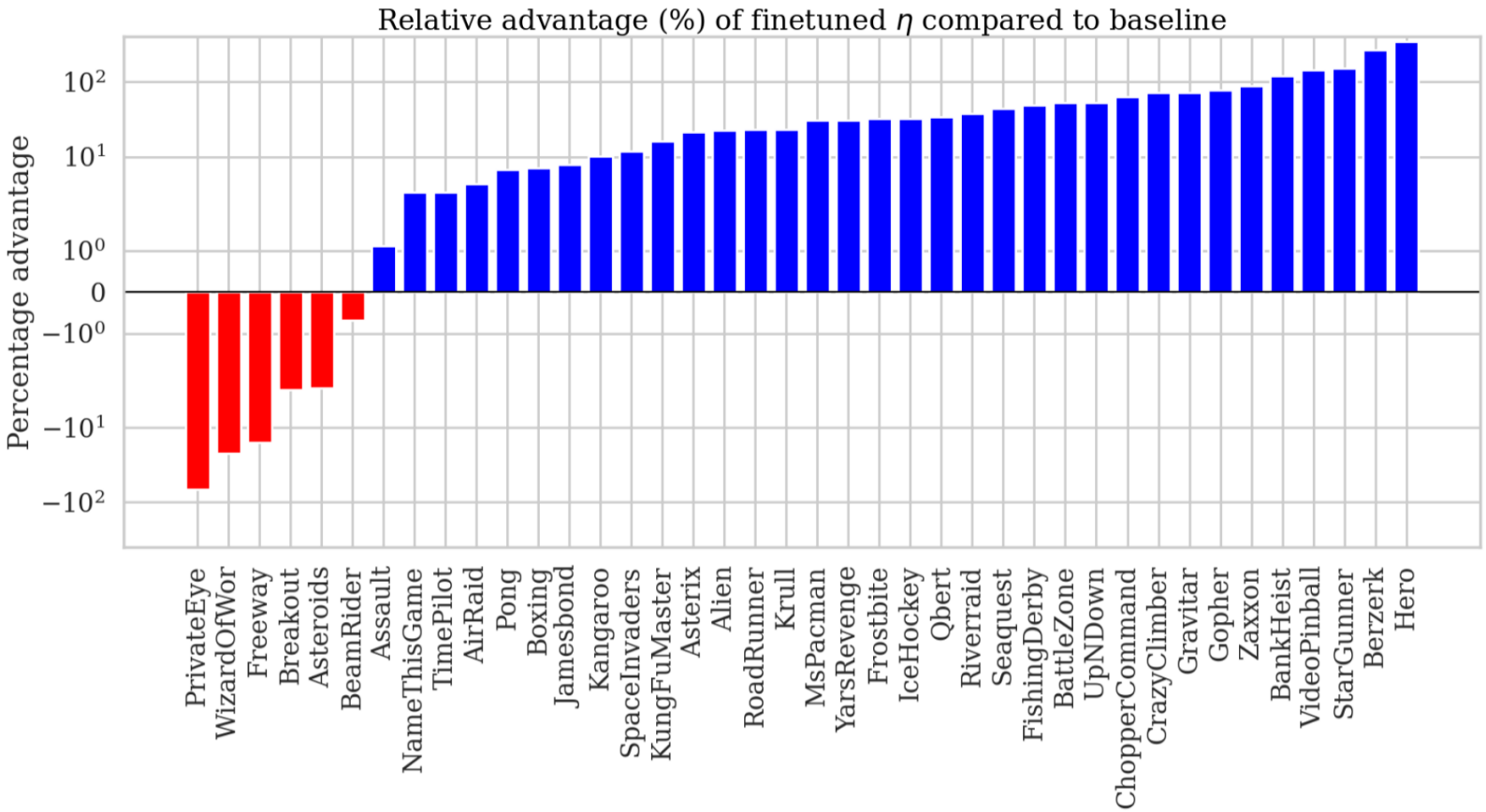
Adaptive potential function

Scaling factor for convergence

Estimate of optimal value function

Reward Shaping

A new dynamical gauge, “self-consistent”:



$$\Phi^{(n-1)}(s) = \eta \max_a Q^{(n-1)}(s,a)$$

$$\bar{r}(s,a,s') = \widetilde{r}(s,a,s') + \gamma\Phi(s') - \Phi(s)$$

$$V^{\infty}(s) = \max_a Q^{\infty}(s,a)$$

$$V^{\infty}(s) = \frac{V_0(s)}{1+\eta}$$

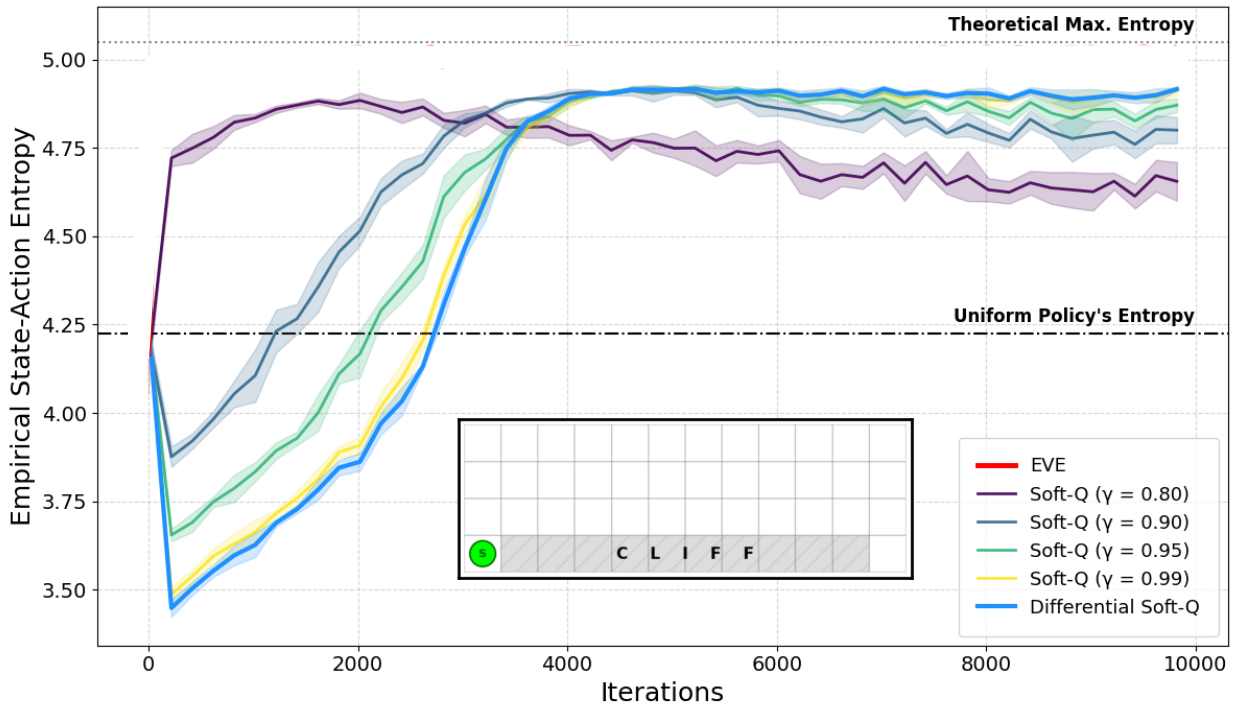
Pure Exploration

Maximum Entropy Exploration Without the Rollouts

UNDER REVIEW

Jacob Adamczyk[1,2], Adam Kamoski[1], Rahul V. Kulkarni[1,2]
{jacob.adamczyk001, adam.kamoski001, rahul.kulkarni}@umb.edu

Visit states uniformly



Novel update equation, sample efficiency

Discovered a new "harmonic" flow function that balances incoming and outgoing transitions to ensure uniform coverage

$$\max_{\pi} \mathbb{H}(d_{p,\pi}) = \max_{\pi} \left(-\sum_{s \in \mathcal{S}} \sum_{a \in \mathcal{A}} d_{p,\pi}(s,a) \log d_{p,\pi}(s,a) \right),$$

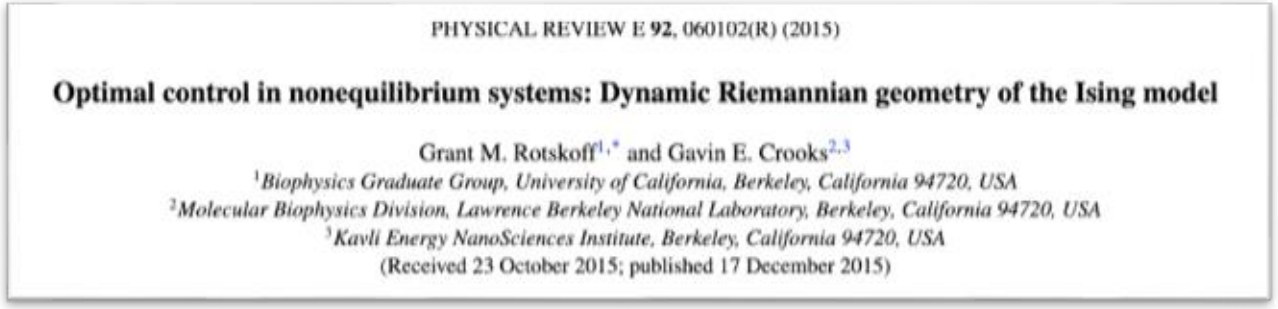
Non-equilibrium

- Define curriculum learning as driving protocol
- Given “A”, how to learn to solve “B”?
  - With minimum work
- Conjugate force is the (centered) feature vector
- Friction tensor emerges as two-point correlator

$$\delta X_i(s,a) = \frac{\partial r_\lambda(\mathbf{s},\mathbf{a})}{\partial \lambda_i} - \mathbb{E}\left[\frac{\partial r_\lambda(\mathbf{s},\mathbf{a})}{\partial \lambda_i}\right]$$

$$\zeta_{ij}(\lambda) = \beta \sum_{t=0}^{\infty} \mathbb{E}_{\tau\sim p_\lambda}(\delta X_i(\mathbf{s}_t,\mathbf{a}_t)\cdot \delta X_j(\mathbf{s}_0,\mathbf{a}_0))$$

$$\mathcal{W}_{\text{excess}} = \int_0^\infty \dot{\lambda}_i(t)\zeta_{ij}(\lambda(t))\dot{\lambda}_j(t)\,dt$$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Optimal control in nonequilibrium systems: Dynamic Riemannian geometry of the Ising model

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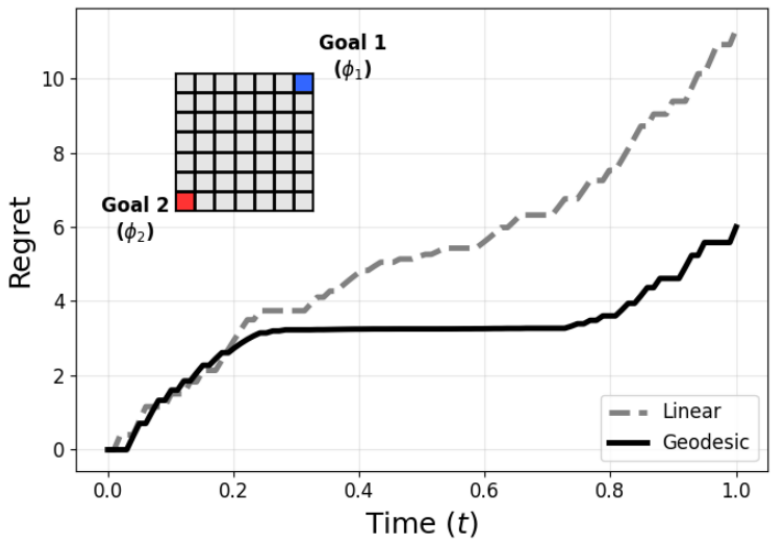
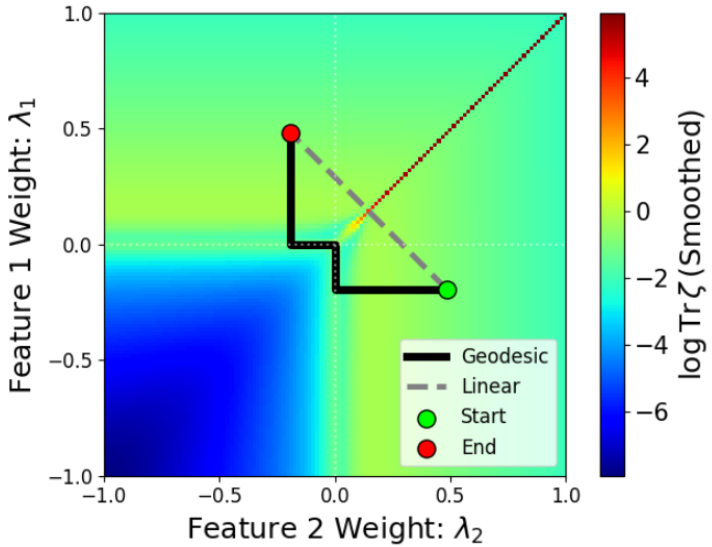
$$\boxed{\ddot{\lambda}^k + \Gamma^k_{ij}(\lambda)\,\dot{\lambda}^i\dot{\lambda}^j = 0}$$

Non-equilibrium

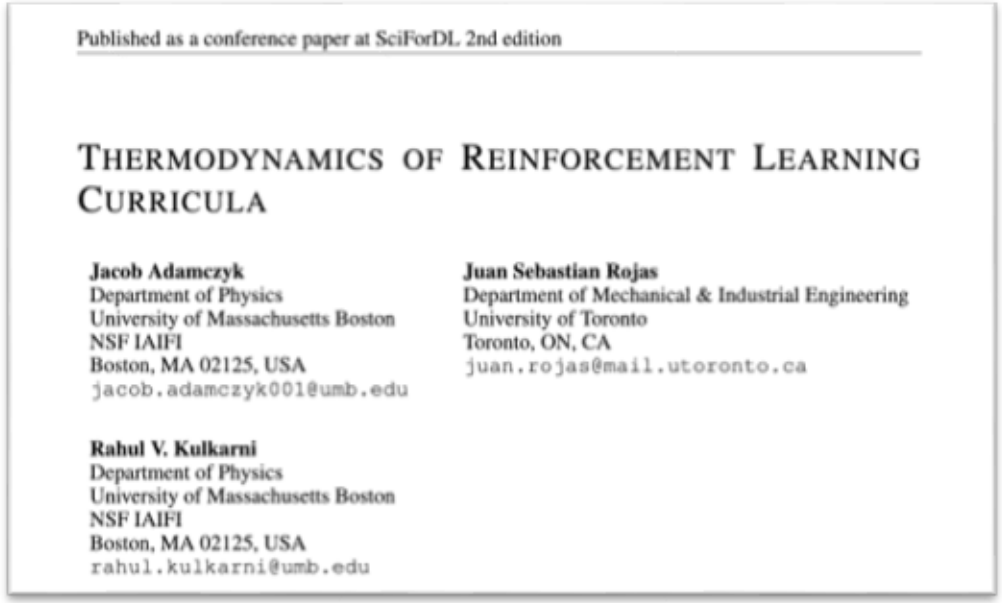
- Define “curriculum learning” as driving protocol
- Use tools from NESM to derive optimality
- Geodesics in task space





Non-equilibrium

- MEW ("Minimum Excess Work")
- To use in deep RL (online learning), control the temperature parameter
  - Type of simulated annealing



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THERMODYNAMICS OF REINFORCEMENT LEARNING CURRICULA

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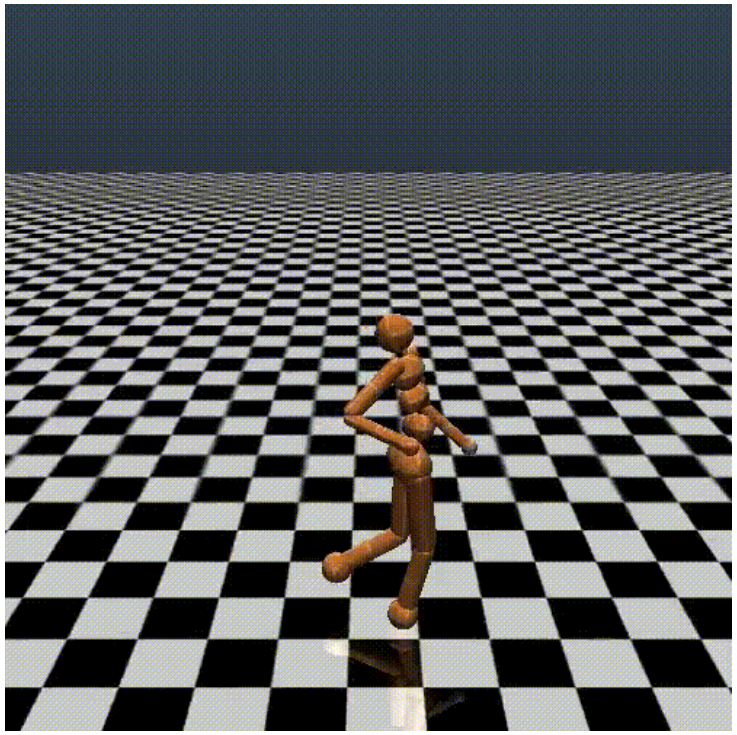

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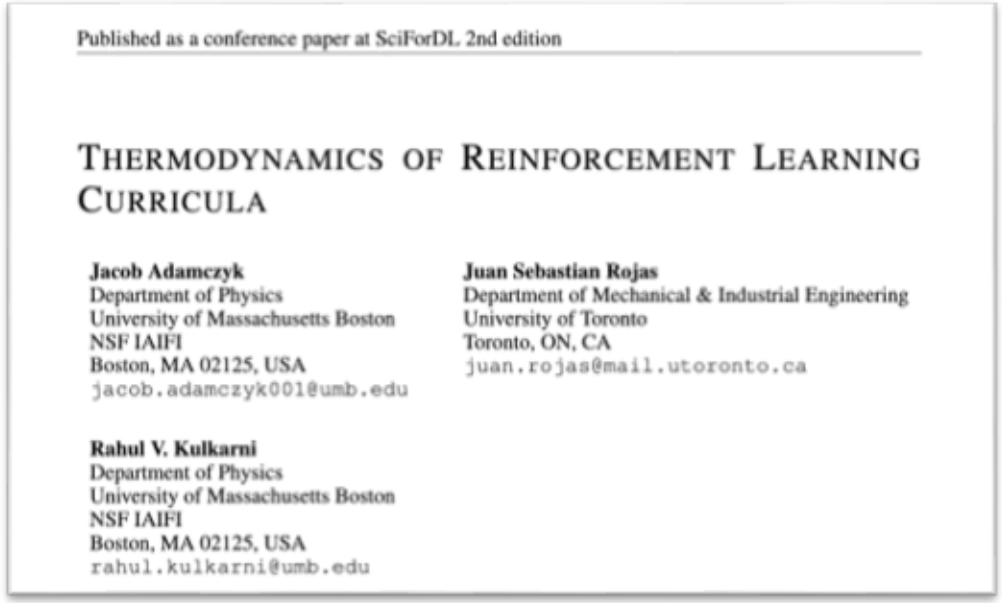

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THERMODYNAMICS OF REINFORCEMENT LEARNING CURRICULA

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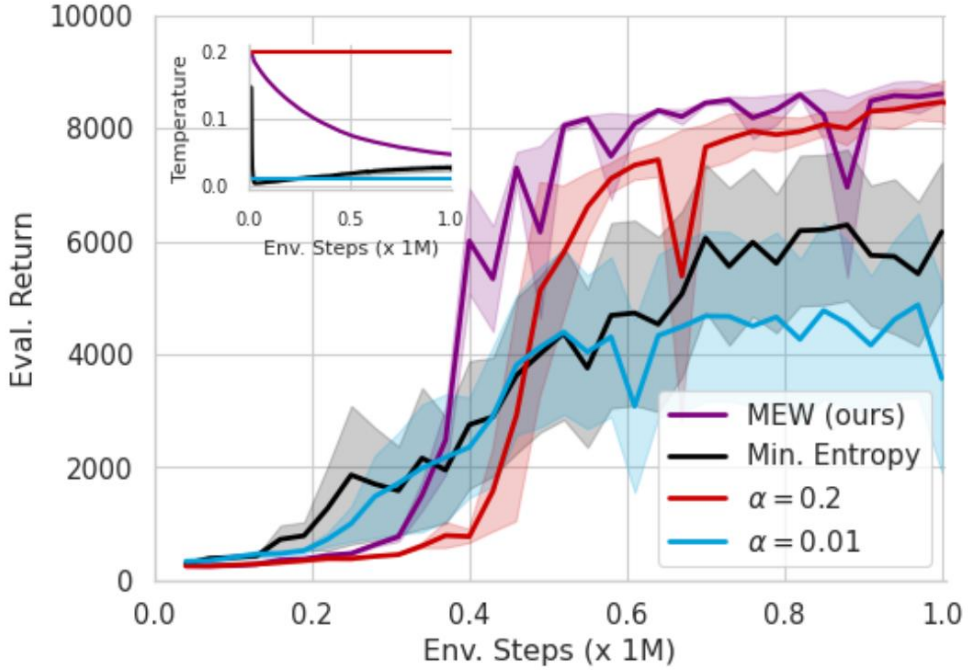

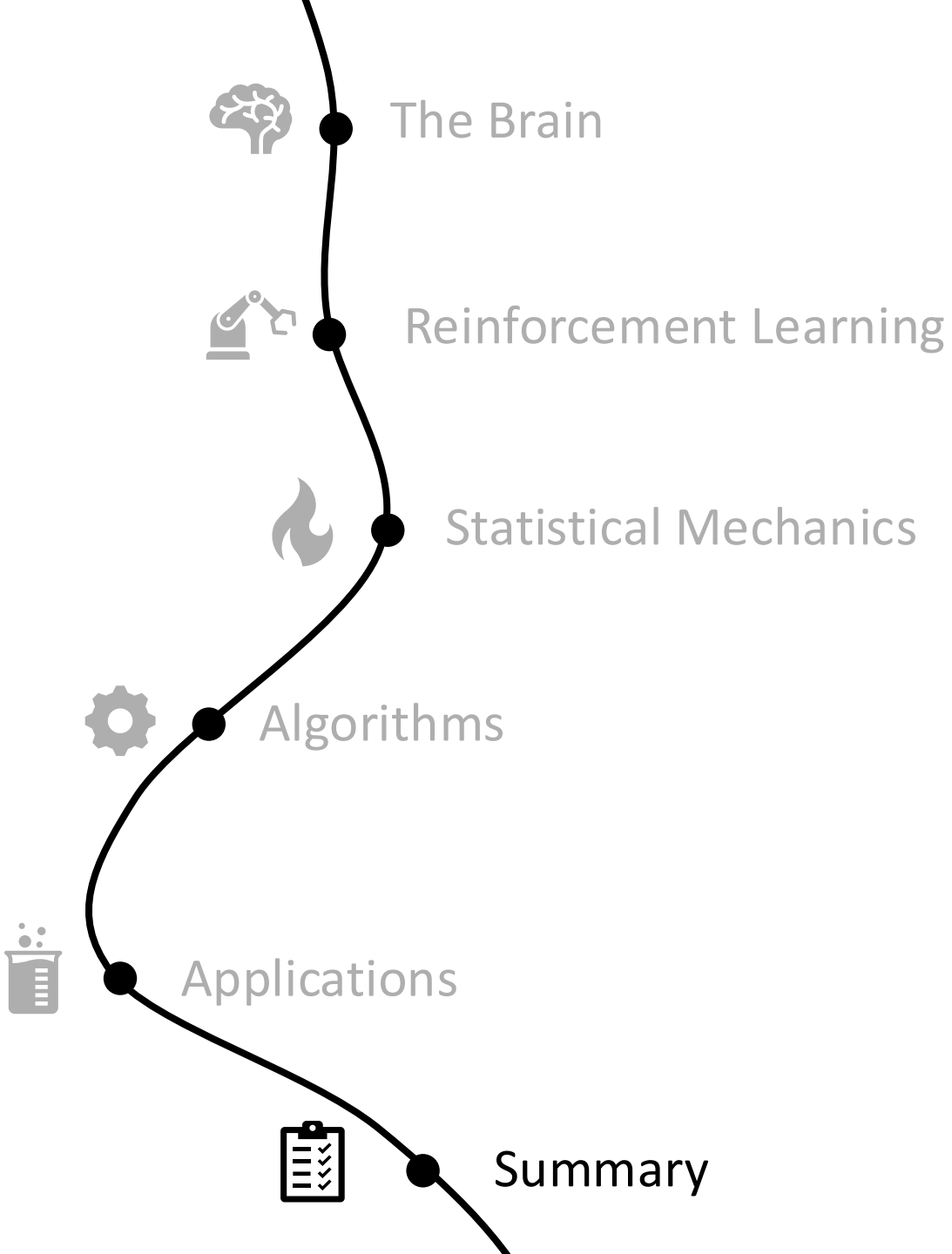
The Brain
Reinforcement Learning
Statistical Mechanics
Algorithms
Applications
Summary

Dictionary

Reinforcement Learning	Statistical Mechanics
Trajectory	Microstate
Reward Function	Hamiltonian
Value Function	Free energy
Regularization Strength	Temperature
Steady-State; Optimal Policy	Thermal Distribution
Reset Penalty	Chemical Potential
Curriculum	Driving Protocol

Summary

- Reinforcement Learning (RL) models bran at low- and high-level
  - Developments in RL have led to advancements in neuroscience, & vice versa
- Established a “stat mech of RL”
  - Trajectory = microstate
  - Ensembles
  - Thermodynamics
- Algorithms that emerge are novel, SOTA
- Applications have just scratched the surface!
  - Invariance, Exploration, Non-equilibrium driving

Thank You!

<u>Collaborators</u>

- James MacGlashan
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- Josiah C. Kratz
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- Gautham Vasan
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